



From Retrieval-Augmented Generation (RAG) to Agentic AI

Building a Local AI-Powered Academic Advisor using Open-Source Large Language Models



Dr. A. A. Nippun Kumaar

Department of Computer Science & Engineering (AI & ML)
Dayananda Sagar University

Problem Statement

Why an AI Academic Advisor is Needed?



-  Curriculum documents are **lengthy**.
-  Students ask **repetitive questions**.
-  Keyword search is **inefficient**.
-  LLMs alone may **hallucinate**.
-  Need **accurate document-grounded responses**.



An AI Academic Advisor that leverages Retrieval-Augmented Generation (RAG) can provide accurate, **document-grounded, and context-aware** academic guidance to students.

01



**Completely Local
AI Assistant**

02



No Cloud APIs

03



**Accurate Curriculum
Retrieval**

04



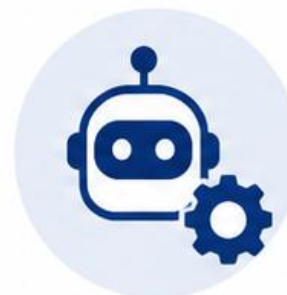
**Reduced
Hallucinations**

05



**Official Document
Grounding**

06



**Foundation for
Agentic AI**





Python

Core programming language
for development



Ollama

Run and manage LLMs
locally



Qwen3

Open-source LLM for
natural language generation



Nomic Embed

Generate high-quality
text embeddings



ChromaDB

Vector database for
similarity search



PyMuPDF

Extract and process
PDF documents



OpenAI Compatible API

Standard API interface
for LLM access

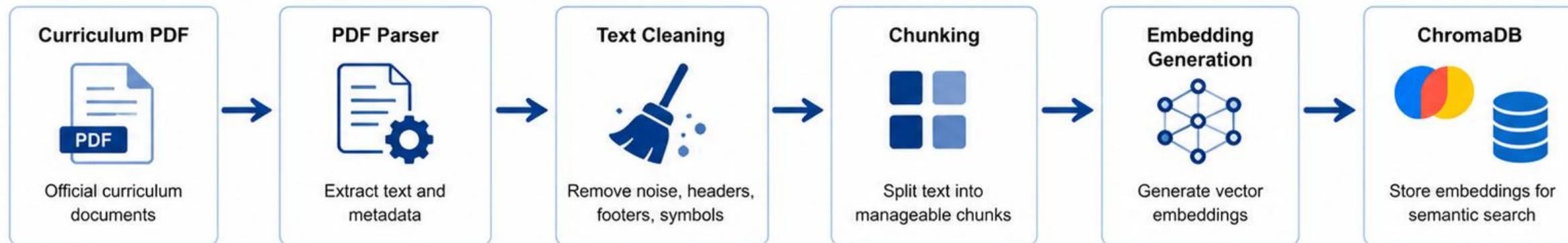


VS Code

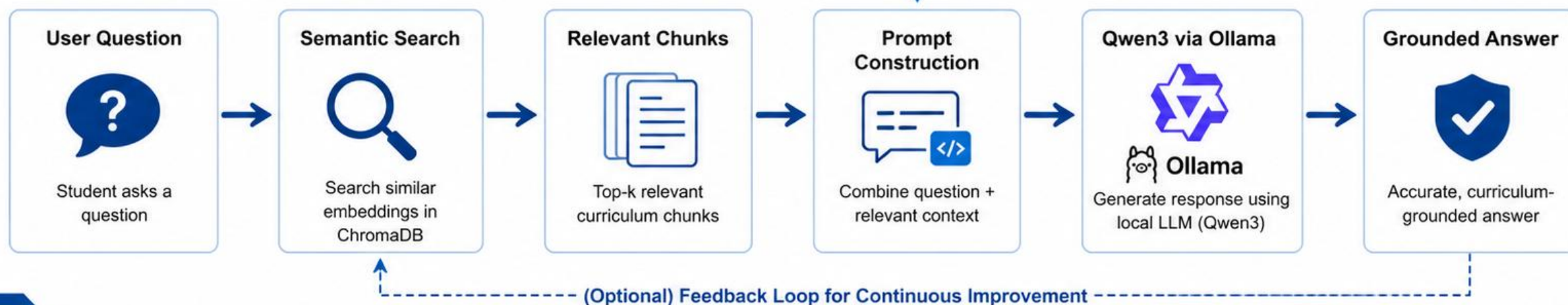
Integrated development
environment



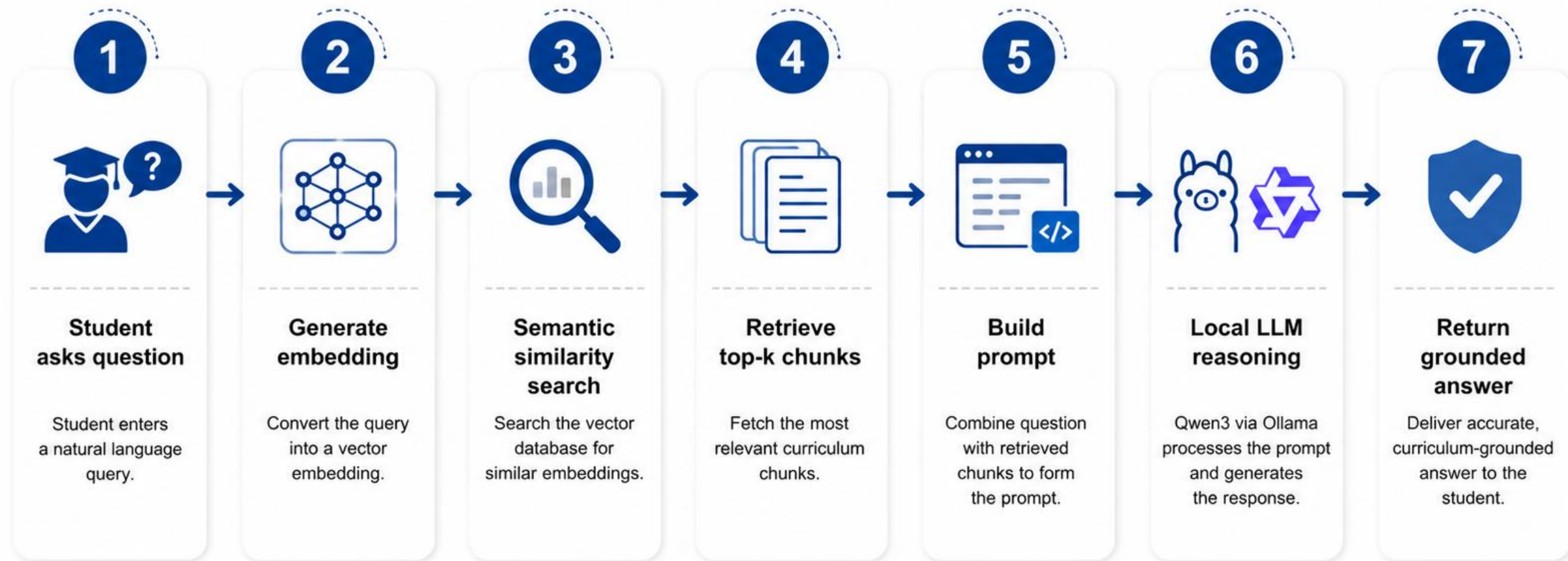
1. Document Ingestion & Indexing Pipeline



2. Query & Retrieval (RAG) Pipeline



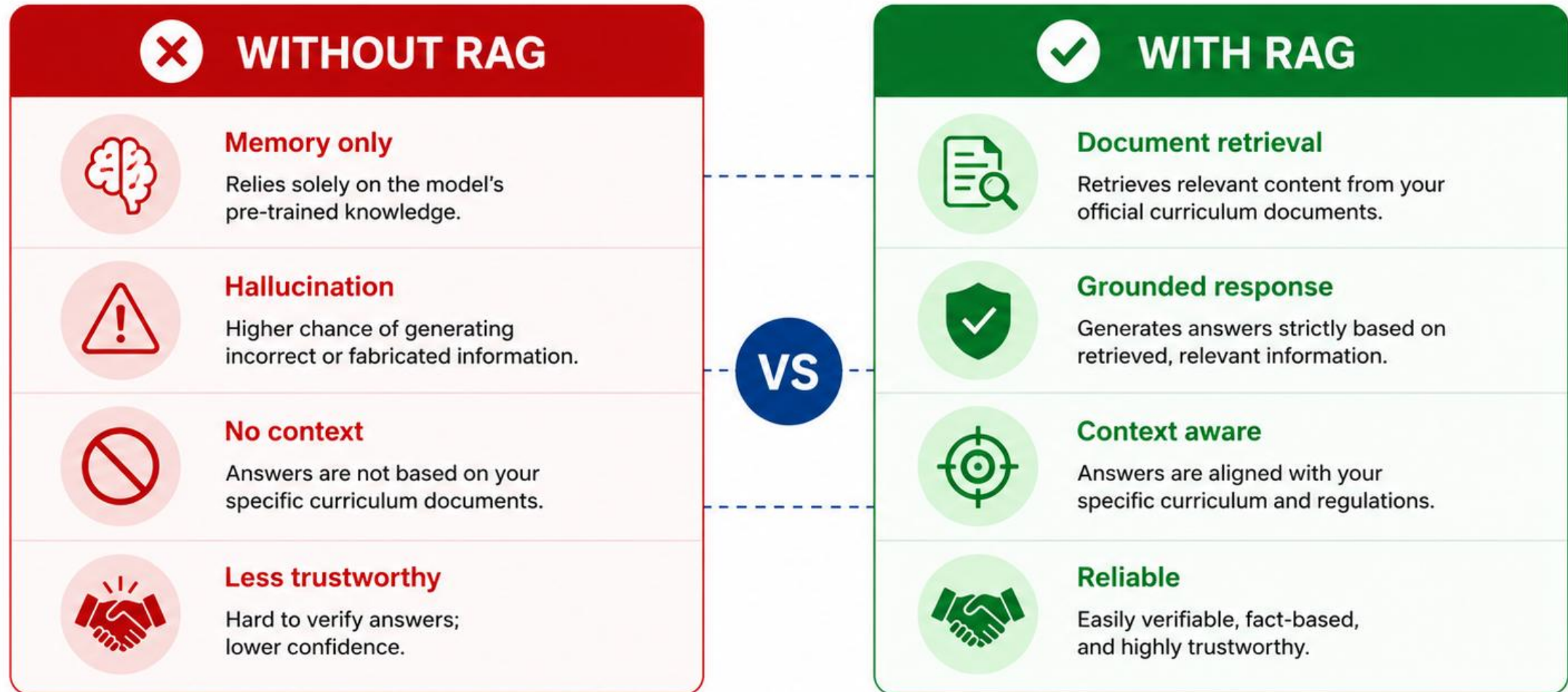
Working Pipeline



All processing is done locally • No data leaves your system • Accurate, private, and reliable answers



Why RAG?



RAG connects the power of LLMs with your official documents for accurate, trustworthy, and context-aware answers.



1 Student asks question



Student

10:15 AM

"What are the objectives of Artificial Intelligence?"



Type your question...



Natural language question entered by the student.

2 System retrieves relevant information



Retrieval Engine

Searching curriculum documents...



AI Syllabus

Artificial Intelligence (Course Code: AI301)

Source: syllabus_ai301.pdf • Page 2



Objectives

Course objectives and intent

Source: syllabus_ai301.pdf • Page 2-3



Course Outcomes

CO1 – CO5 defined

Source: syllabus_ai301.pdf • Page 3



Module Details

Module-wise topics and hours

Source: syllabus_ai301.pdf • Page 4-6



Top-k relevant chunks retrieved from the vector database (ChromaDB).

3 Grounded AI response



AI Academic Advisor

10:16 AM



According to the official AI syllabus, the objectives of the course "Artificial Intelligence" are:

- 1 To introduce the fundamental concepts and techniques of Artificial Intelligence.
- 2 To understand problem solving using intelligent algorithms and reasoning.
- 3 To apply AI techniques to real-world problems through practical examples.
- 4 To develop skills in designing and implementing intelligent systems.

These objectives are defined in the official syllabus document.

Source: syllabus_ai301.pdf (Page 2-3)



Answer generated by Qwen3 via Ollama, grounded in retrieved curriculum documents.



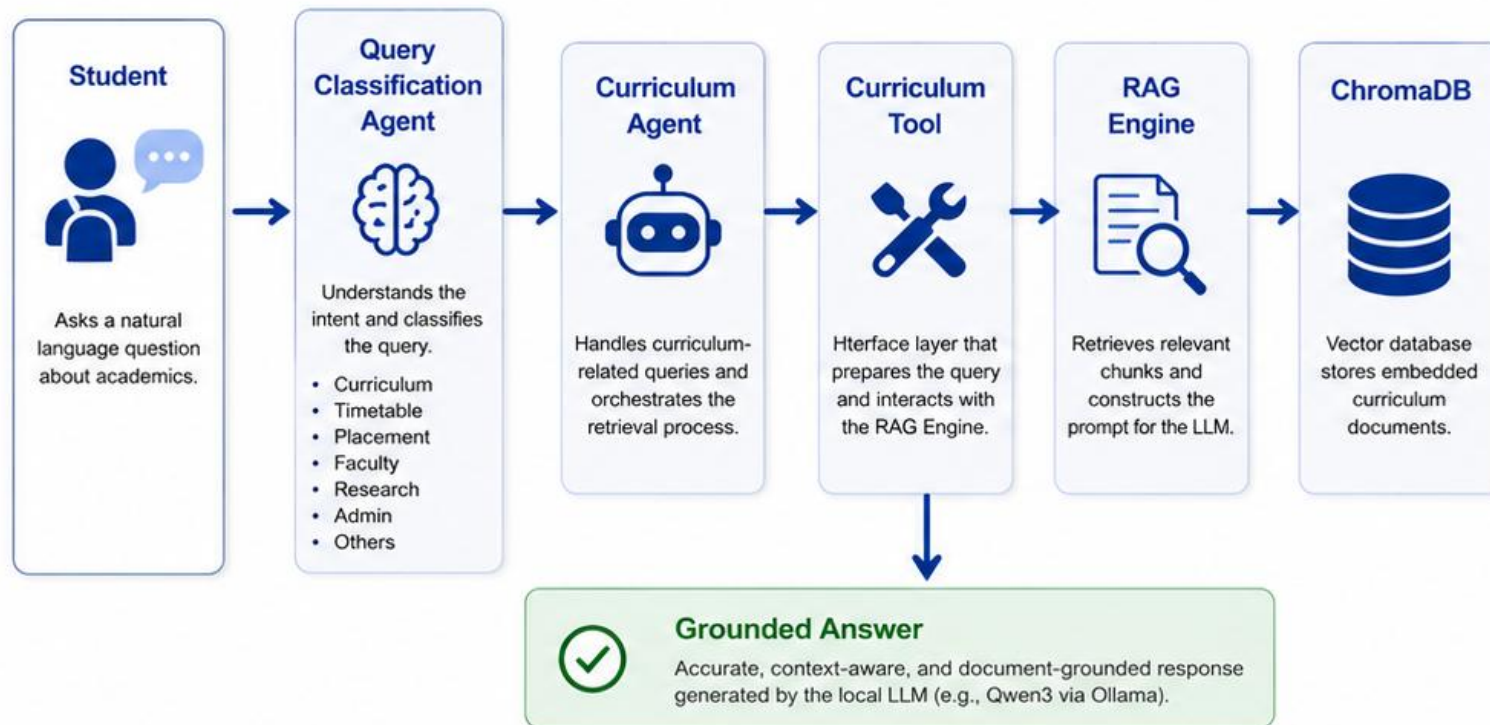
Accurate. Contextual. Document-Grounded. 100% Local.



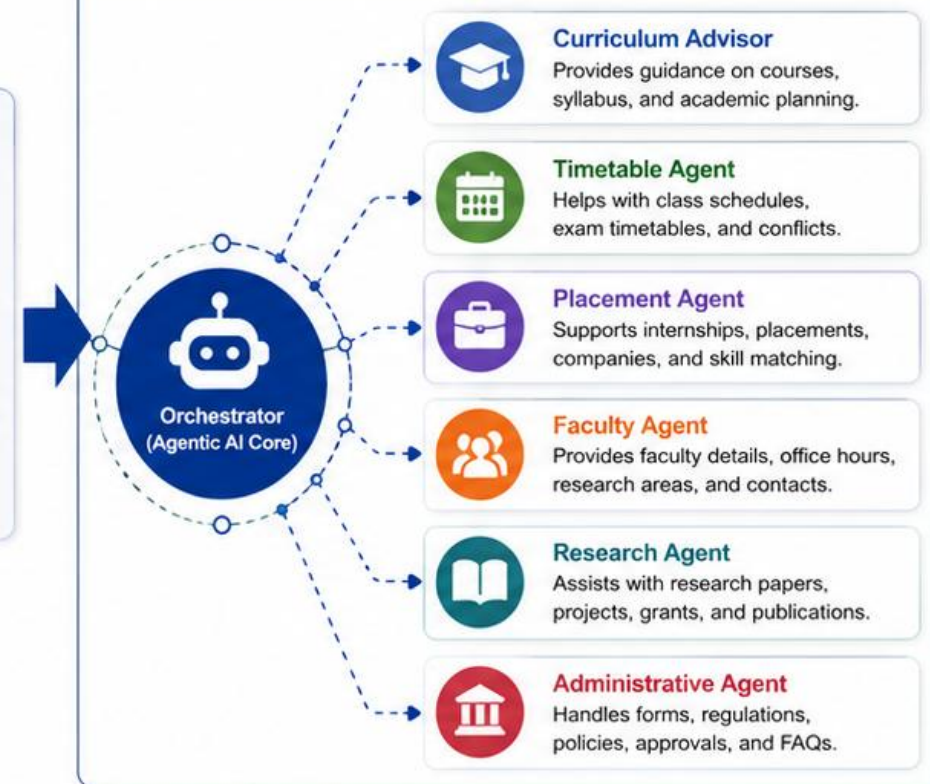
Towards Agentic AI

From a Single-Purpose RAG System to a Multi-Agent Academic Ecosystem

CURRENT RAG ARCHITECTURE



FUTURE: MULTI-AGENT ACADEMIC ECOSYSTEM



Specialized Agents
for Every Academic Need



Intelligent Orchestration
& Collaboration



Higher Accuracy
& Relevance



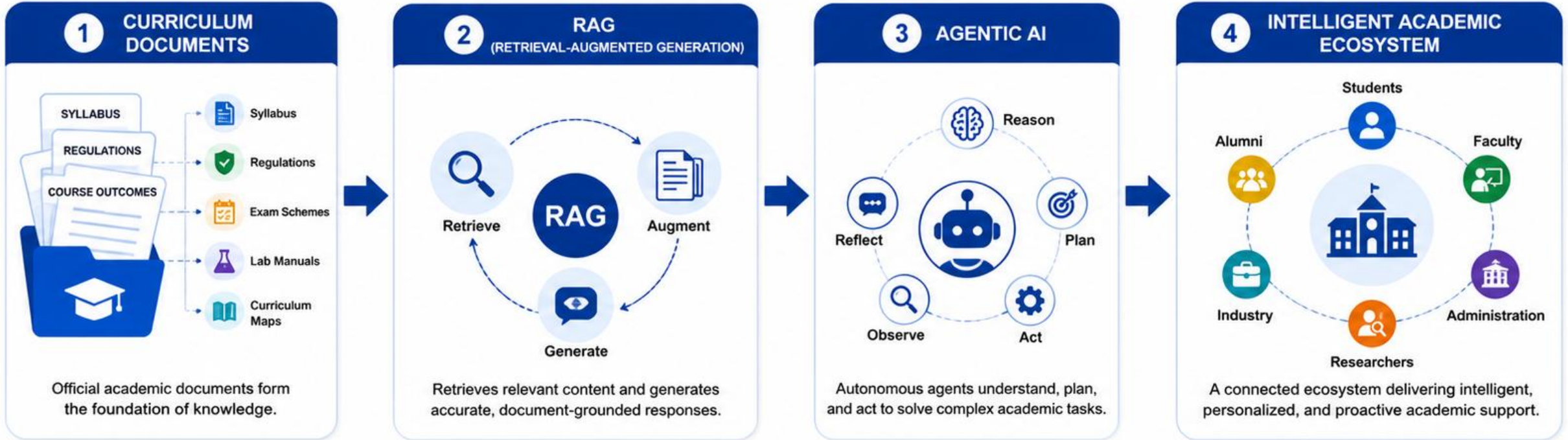
100% Local, Secure
& Private



Scalable for Future
Academic Use Cases



From Curriculum Documents to an Intelligent Academic Ecosystem



ACCURATE

Document-grounded responses ensure high accuracy.



LOCAL

Runs completely on local infrastructure. No cloud dependency.



RELIABLE

Reduces hallucinations. Consistent and trustworthy answers.



PRIVATE

Your data stays in your system. 100% privacy.



READY FOR AGENTIC AI

A strong foundation for building specialized AI agents and workflows.



Building the Future of Academic Support — Intelligent, Private, and Trusted.

